Project Proposal

Goal
To develop an agentic AI app that provides personalized travel destination recommendations based on user preferences, needs, budget, and interests using Google Opal.

Problem Statement & Motivation
Individuals often have a hard time choosing a travel destination that fits their needs due to the vast amount of options and searching it takes to find the right place. Travelers need to consider factors such as their budget, weather, the travel distance, activities provided, and seasonal availability. Even though travel websites offer travel plans and suggestions, it often lacks detailed planning and personalization to an individuals preferences.

The problem addressed in this project is the lack of intelligent systems that can analyze user preferences and work to recommend travel destinations tailored to one's needs and likes. Oftentimes, travelers spend upwards of 5 hours researching a travel destination, which can be a waste of time and lead to decision fatigue and inefficient planning.

The target users for this application include:

- Individuals or couples unsure of where they would like to travel.
- Students or travelers with a tight budget looking for affordable destinations.
- Frequent travelers seeking new and unique places to explore.
- Families or friends looking to enjoy a vacation suited to their needs.

With this, an agentic AI app would be the solution to this problem. Creating a destination recommendation requires collecting user preferences and travel accommodations through a series of asking questions. Agent based systems allow different AI systems to work with one another, such as data retrieval, preference analysis, and evaluation. This multi agent system enables flexibility and further presonalizes travel destination recommendations that cater to each individuals needs.

Use Cases & Scenarios

Scenario 1: Destination Discovery based on Interest

A user want to recieve travel destination suggestions based on what they like. They are interested in vacationing near a beach, while also exprerinceing the local culture and enjoying the food.

Agent tasks:

- The preference analysis agent collects and interprets user interests and travel preferences.
- The data fetcher agent gathers information about potential destinations that match user interests and preferences.
- The recommendation agents compiles a list of personalzed travel destinaitons.
- The summarizer agent presents destination options and a reason for its recommendation.

Scenario 2: Destination Recommandtion based on Budget

A user wants to explore travel destinations that fit within their budget range.

Agent tasks:

- Preference analysis agent gathers user budget and travel constraints.
- The data fetcher agent retrieves average travel costs and seasonal pricing data.
- The recommendation agent suggests destinations that fit budget requirements and preferences.
- The critic agent ensures recommendations fit budget needs and user wants.

Scenario 3: Seasonal Travel Recommendation

A user wants to know the best travel destinations for a specific season.

Agent tasks:

- The preference analysis agent gathers preferred travel dates and climate preferences.
- The data fetcher agent retrieves seasonal weather patterns and peak travel periods.
- The recommendation agent suggests destinations that match seasonal suitability.
- The summarizaer agent explains why each destination is recommended during that season.

Initial System Design

The system will consist of multiple AI agents implemented through Google Opal's agentic framework.

Proposed Agents and Roles

1. Prefernce Analysis Agent
   - Collects and analyses user interests, budget, travel style, and constraints.
2. Data Fetcher Agent
   - Retrieves travel-related data including destination details, weather, cost estimates, and activity availability.
3. Recommendation Agent
   - Generates personalized destination suggestions using user preferences and retrieved travel data.
4. Summarizer Agent
   - Produces clear descriptions explaining why each destination is recommended.
5. Critic Agent
   - Evaluates recommendations for accuracy, relevance and feasibility.

Feasability & Risks

Potential Challenges:

- One challenge is AI hallucination, where the system can generate inaccurate or outdated information. This risk can be reduced by using verified API's and ensuring validation through the Critic Agent.
- There could also be a challenge in ensuring recommendations continue to stay relevant when user preferences are unclear or not easy to decipher.

Constraints:
- API rate limits
- Privacy concerns regarding storage of user preferences.
- Limitations within Google Opal

Timeline for Project Execution

Week 1-2:
- Finalize system design and identify required APIs and data sources.

Week 3-4:
- Develop preference analysis agent and data fetcher agent.

Week 5-6:
- Develop recommendation agent and summarizer agent.
- Integrate external travel and weather API's.

Week 7:
- Implement critic agent
- Conduct system testing and validation.

Week 8:
- Preform user testing and system refinement.
- Prepare final documentation and presentation.